

DBMS – 16 Mark Questions (Normalization)

Based on the uploaded notes. [filecite](#) [turn0file0](#)

1. Explain each of the following normal forms with example.

Normalization is the process of organizing data in a database to reduce redundancy and avoid anomalies such as insertion, deletion, and update anomalies. It divides large tables into smaller related tables and establishes relationships between them.

The main objectives of normalization are:

- To reduce data redundancy
 - To improve data consistency
 - To eliminate anomalies
 - To maintain data integrity
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(i) First Normal Form (1NF)

Definition

A relation is said to be in First Normal Form (1NF) if:

- All attributes contain only atomic (single) values.
- Repeating groups or multivalued attributes are not allowed.
- Each column contains values of a single type.

In simple words, each cell in the table should contain only one value.

Example

Table NOT in 1NF

StudentID	StudentName	PhoneNumbers
S101	Ravi	9876543210, 9123456780
S102	Priya	9988776655

Problem:

- The attribute PhoneNumbers contains multiple values in a single cell.
 - Hence the table violates 1NF.
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Conversion into 1NF

StudentID	StudentName	PhoneNumber
S101	Ravi	9876543210
S101	Ravi	9123456780
S102	Priya	9988776655

Now:

- Every attribute contains atomic values.
 - Repeating groups are removed.
 - Therefore the table is in 1NF.
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Advantages of 1NF

- Eliminates repeating groups
 - Simplifies querying
 - Improves data organization
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(ii) Second Normal Form (2NF)

Definition

A relation is in Second Normal Form (2NF) if: 1. It is already in 1NF. 2. Every non-prime attribute is fully functionally dependent on the whole primary key. 3. Partial dependency should not exist.

Partial dependency occurs when a non-key attribute depends on only part of a composite key.

Example

Consider the table:

StudentID	CourseID	StudentName	CourseName	Marks
S101	C1	Ravi	DBMS	85
S101	C2	Ravi	Java	90
S102	C1	Priya	DBMS	88

Primary Key = (StudentID, CourseID)

Functional Dependencies:

- StudentID $\rightarrow$ StudentName
- CourseID $\rightarrow$ CourseName
- (StudentID, CourseID) $\rightarrow$ Marks

Problem:

- StudentName depends only on StudentID.
- CourseName depends only on CourseID.
- These are partial dependencies.

Hence the table is not in 2NF.

Decomposition into 2NF

Student Table

StudentID	StudentName
S101	Ravi
S102	Priya

Course Table

CourseID	CourseName
C1	DBMS
C2	Java

Marks Table

StudentID	CourseID	Marks
S101	C1	85

StudentID	CourseID	Marks
S101	C2	90
S102	C1	88

Now:

- Every non-key attribute depends on the entire primary key.
- Partial dependency is removed.
- Hence the relation is in 2NF.

Advantages of 2NF

- Reduces redundancy
- Eliminates partial dependency
- Improves consistency

(iii) Third Normal Form (3NF)

Definition

A relation is in Third Normal Form (3NF) if: 1. It is already in 2NF. 2. There should be no transitive dependency.

Transitive dependency occurs when a non-key attribute depends on another non-key attribute.

Example

EmpID	EmpName	DeptID	DeptName
E101	Arun	D1	Sales
E102	Meena	D2	HR

Functional Dependencies:

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}$

Problem:

- DeptName depends on DeptID.

- DeptID depends on EmpID.
- Therefore DeptName transitively depends on EmpID.

Hence the relation is not in 3NF.

Decomposition into 3NF

Employee Table

EmpID	EmpName	DeptID
E101	Arun	D1
E102	Meena	D2

Department Table

DeptID	DeptName
D1	Sales
D2	HR

Now:

- Transitive dependency is removed.
 - Non-key attributes depend only on the primary key.
 - Hence the relation is in 3NF.
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Advantages of 3NF

- Eliminates transitive dependency
 - Reduces duplication
 - Improves integrity
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(iv) Boyce-Codd Normal Form (BCNF)

Definition

A relation is in Boyce-Codd Normal Form (BCNF) if:

- For every functional dependency $X \rightarrow Y$, X must be a super key.

BCNF is a stronger version of 3NF.

Example

Consider the relation:

Student	Subject	Teacher
Ravi	DBMS	Kumar
Priya	Java	Arun
Ravi	Java	Arun

Functional Dependencies:

- $(\text{Student}, \text{Subject}) \rightarrow \text{Teacher}$
- $\text{Teacher} \rightarrow \text{Subject}$

Candidate Keys:

- $(\text{Student}, \text{Subject})$
- $(\text{Student}, \text{Teacher})$

Problem:

- Teacher determines Subject.
 - Teacher is not a super key.
 - Hence BCNF is violated.
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Decomposition into BCNF

Teacher Table

Teacher	Subject
Kumar	DBMS
Arun	Java

StudentTeacher Table

Student	Teacher
Ravi	Kumar

Student	Teacher
Priya	Arun
Ravi	Arun

Now:

- Every determinant is a candidate key or super key.
- Therefore the relation is in BCNF.

Advantages of BCNF

- Removes redundancy completely
- Eliminates anomalies
- Provides better consistency

2. Demonstrate the usage of the following Normal forms in eliminating redundancy and anomalies.

(i) Fourth Normal Form (4NF)

Definition

A relation is in Fourth Normal Form (4NF) if: 1. It is already in BCNF. 2. It should not contain non-trivial multivalued dependencies.

4NF deals with multivalued dependencies.

Multivalued Dependency (MVD)

A multivalued dependency occurs when:

- One attribute determines multiple independent values of another attribute.

Notation:

$A \twoheadrightarrow B$

Meaning:

- For one value of A, there may be multiple independent values of B.
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Example

Student	Hobby	Language
Ravi	Cricket	English
Ravi	Cricket	Hindi
Ravi	Music	English
Ravi	Music	Hindi

Here:

- A student can have many hobbies.
- A student can know many languages.
- Hobbies and languages are independent.

MVDs:

- Student $\twoheadrightarrow$ Hobby
- Student $\twoheadrightarrow$ Language

Problems:

- Large redundancy
- Repeated data
- Update anomalies

Hence the table is not in 4NF.

Decomposition into 4NF

StudentHobby Table

Student	Hobby
Ravi	Cricket
Ravi	Music

StudentLanguage Table

Student	Language
Ravi	English
Ravi	Hindi

Now:

- Multivalued dependencies are removed.
- Redundancy is minimized.
- Therefore the relation is in 4NF.

Advantages of 4NF

- Eliminates multivalued dependency
- Reduces repeated data
- Removes update anomalies

(ii) Multivalued Dependency (MVD)

Definition

A multivalued dependency exists when:

- One attribute in a table uniquely determines multiple values of another attribute, independently of other attributes.

Notation:

$A \twoheadrightarrow B$

This means:

- For a single value of A, there are multiple values of B.

Example

Employee	Skill	Project
Arun	Java	Banking

Employee	Skill	Project
Arun	Java	Insurance
Arun	Python	Banking
Arun	Python	Insurance

Here:

- An employee may have multiple skills.
- An employee may work on multiple projects.
- Skills and projects are independent.

MVDs:

- Employee → Skill
- Employee → Project

Problems caused by MVD:

- Data redundancy
- Insertion anomaly
- Deletion anomaly
- Update anomaly

Solution

Decompose into:

EmployeeSkill

Employee	Skill
Arun	Java
Arun	Python

EmployeeProject

Employee	Project
Arun	Banking
Arun	Insurance

This removes unnecessary duplication.

(iii) Fifth Normal Form (5NF)

Definition

A relation is in Fifth Normal Form (5NF) if: 1. It is already in 4NF. 2. It cannot be decomposed further without losing information. 3. Every join dependency is implied by candidate keys.

5NF is also called Project-Join Normal Form (PJNF).

Example

Consider:

Supplier	Part	Project
S1	P1	J1
S1	P2	J1
S2	P1	J2

Suppose:

- Suppliers supply parts.
- Parts are used in projects.
- Suppliers work for projects.

The relation may be decomposed into:

SupplierPart

Supplier	Part
S1	P1
S1	P2
S2	P1

PartProject

Part	Project
P1	J1
P2	J1

Part	Project
P1	J2

SupplierProject

Supplier	Project
S1	J1
S2	J2

Joining these tables reconstructs the original relation.

Advantages of 5NF

- Eliminates join anomalies
- Avoids redundant decompositions
- Ensures lossless joins

(iv) Join Dependency and Lossless Decomposition

Join Dependency (JD)

Definition

A join dependency exists when:

- A table can be decomposed into multiple tables and reconstructed by joining them without losing information.

It is represented as:

JD(R1, R2, R3, ...)

Example

Consider the relation:

Student	Course	Faculty
Ravi	DBMS	Kumar

Student	Course	Faculty
Priya	Java	Arun

It can be decomposed into:

StudentCourse

Student	Course
Ravi	DBMS
Priya	Java

CourseFaculty

Course	Faculty
DBMS	Kumar
Java	Arun

When these tables are joined, the original relation is obtained.

This is called Join Dependency.

Lossless Decomposition

Definition

A decomposition is said to be lossless if:

- After decomposing a relation into smaller relations, joining them again gives exactly the original relation without generating extra tuples.

Example

Original Relation:

RollNo	Name	Dept
101	Ravi	CSE
102	Priya	IT

Decompose into:

Student

RollNo	Name
101	Ravi
102	Priya

Department

RollNo	Dept
101	CSE
102	IT

Joining Student and Department tables produces the exact original relation.

Hence it is a lossless decomposition.

Advantages of Lossless Decomposition

- Prevents information loss
- Maintains consistency
- Improves database design

Conclusion

Normalization is an important technique in DBMS used to organize data efficiently. Different normal forms such as 1NF, 2NF, 3NF, BCNF, 4NF, and 5NF help eliminate redundancy and anomalies. Concepts like multivalued dependency, join dependency, and lossless decomposition ensure efficient and reliable database design.

Proper normalization:

- Reduces redundancy
- Eliminates anomalies
- Improves consistency
- Enhances database performance and integrity